DPP- 01

CS & IT

Algorithm

Heap Algorithms

Q1 Which of the following is valid min heap?
(A) 34, 38, 35, 39, 40, 36, 50
(B) 25, 14, 12, 13, 10, 8, 16
(C) Both (a) and (b)
(D) Neither (a) nor (b)

Q2 The minimum number of interchanges needed to convert the array into a max-heap is
89, 19, 40, 17, 12, 10, 2, 5, 7, 11, 6, 9, 70
(A) 0 (B) 1
(C) 2 (D) 3

Q3 Consider a complete binary tree where the left and right subtrees of the root are max -heaps. The lower bounds for the number of operations to convert the tree to a heap is
(A) $\Omega(\log n)$ (B) $\Omega(1)$
(C) $\Omega(n \log n)$ (D) $\Omega(n^2)$

Q4 Consider a max heap represented by the array :
40, 30, 20, 10, 15, 16, 17, 8, 4

Array	1	2	3	4	5	6	7	8	9
Value	40	30	20	10	15	16	17	8	4

Now consider that a value 35 is inserted into the heap. After insertion new heap is : -
(A) 40, 30, 20, 10, 15, 16, 17, 8, 4, 35
(B) 40, 35, 20, 10, 30, 16, 17, 8, 4, 15
(C) 40, 30, 20, 10, 35, 16, 17, 8, 4, 15
(D) 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

Q5 The number of possible min -heap containing each value from {1, 2, 3, 4, 5, 6, 7} exactly once.

Q6 Consider the following statements : -
I. The smallest element in a max heap is always at a leaf node.
II. The second largest element in a max heap is always a child of the root node when Max heap tree contain unique elements.
III. A max-heap can be constructed from a binary search tree in O(n) time.
IV. A binary search tree can be constructed from a max-heap in O(n) time.
Which of the above statements are TRUE?
(A) I, III and IV (B) II, III and IV
(C) I, II and III (D) I, II and IV

Q7 Let H be a binary min-heap consisting of n elements implemented as an array. What is the worst case time complexity of an optimal algorithm to find the maximum element in H?
(A) $\theta(\log n)$
(B) $\theta(n \log n)$
(C) $\theta(n)$
(D) $\theta(1)$

Q8 Which of the following is/are correct?
(A) Bubble sort is stable but not inplace sorting technique
(B) Insertion sort is stable sorting technique.
(C) Selection sort is a stable sorting technique.
(D) Bubble sort is a inplace sorting technique

Q9 Consider the following array

70	60	20	50	40	5	19	21

To sort this array using selection sort what is the result after 3rd pass?
(A) 70, 60, 50, 40, 20, 5, 19, 21
(B) 5, 19, 20, 21, 40, 70, 60, 50

(C) 5, 20, 19, 50, 40, 70, 60, 21
(D) 5, 19, 20, 50, 40, 70, 60, 21

Q10 Consider the following array:

50	70	13	9	85	21	16

How many inversions are present for above array ?

Q11 Consider the following array:

50	70	13	9	85	21	16

How many comparisons are needed to sort the above array using insertion sort?

Answer Key

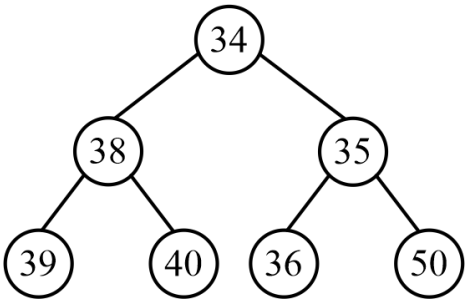
Q1	A
Q2	C
Q3	B
Q4	B
Q5	80
Q6	C
Q7	C
Q8	B, D
Q9	D
Q10	12
Q11	16

Hints & Solutions

Note: scan the QR code to watch video solution

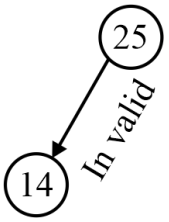
Q1 Text Solution:

(a)

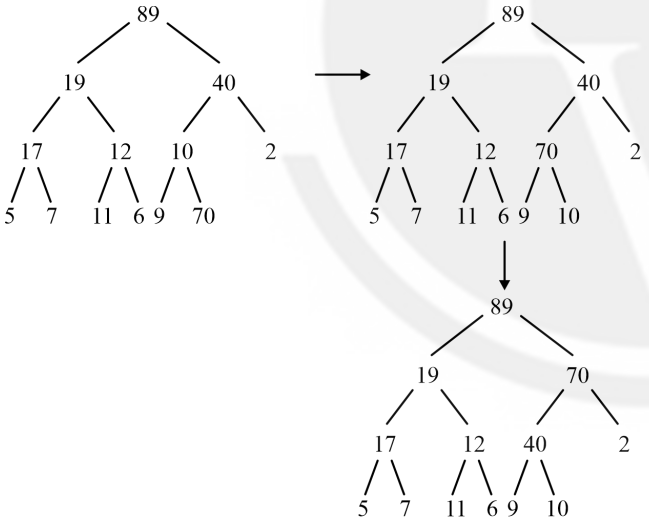


So option a) is valid.

(b)



Q2 Text Solution:



Therefore 2 swaps are need to convert into max heap.

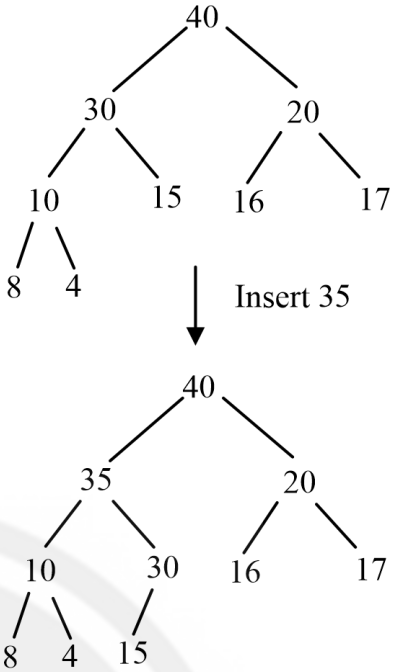
Option (c) is correct.

Q3 Text Solution:

For adjusting the root node in best case we just have to compare with one node thus it will take constant time.

Lower bond for operations is $\Omega(1)$.

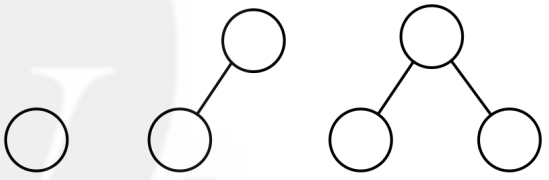
Q4 Text Solution:



40, 35, 20, 10, 30, 16, 17, 8, 4, 15

Option (b) is correct.

Q5 Text Solution:



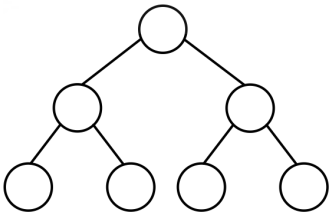
T(1) = 1 T(2) = 1 T(3) = 2

For number of min heaps :-

$$\boxed{T(n)=\sum_{k=0}^{n-1} {}^{n-1}C_k \cdot T(k)\cdot T(n-k-1)}$$

Where, k = number of elements in left subtree.

For '7' elements



3-elements in left and 3 elements in right.

k = 3

$T(7) = {}^{6}C_3\, T(3) \cdot T(7-3-1)$

$= {}^{6}C_3\, T(3) \cdot T(3)$

$= 20 \cdot 4$

$= 80$

Q6 Text Solution:

Binary search tree can not be constructed from max heap in O(n). This statement is only false because it take O(nlogn) time.

Option (c) is correct.

Q7 Text Solution:

Let 'H' be a binary min heap.

To find maximum element we have to look at leaf nodes.

Almost there are (n/2) leaf nodes.

Time complexity will be O(n).

Alternative Approach: -

Convert min heap to max heap root element will be maximum element. This conversion is done with heapify

method and it takes O(n) time.

option (c) is correct.

Q8 Text Solution:

- Bubble sort is stable but not inplace sorting technique **False**
- Insertion sort is stable sorting technique.

True

- Selection sort is a stable sorting technique.

False

- Bubble sort is a inplace sorting technique

True

Q9 Text Solution:

Take first minimum and swap it to the first place.

Result after 3rd pass = 5, 19, 20, 50, 40, 70, 60, 21

Q10 Text Solution:

if i < j && a[i] > a[j]

50 = 13, 9, 21, 16, = 4

70 = 13, 9, 21, 16 = 4

13 = 9 = 1

9 = = 0

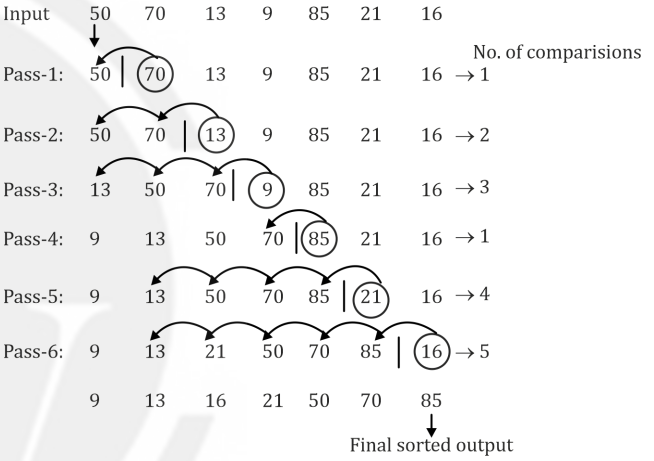
85 = 21,16 = 2

21 =16 = 1

16 = = 0

Number of inversion = 12

Q11 Text Solution:



Total number of comparisons

= 1 + 2 + 3 + 1 + 4 + 5

= 6 + 10 = 16